

Aluminum electrolytic capacitors

Snap-in capacitors

Series/Type: **B43657**

Date: February 2026

Snap-in capacitors

B43657

Ultra compact – 105 °C

General-purpose grade capacitors

Applications

- Power supplies
- Frequency converters
- Uninterruptible power supplies
- Medical appliances
- Solar inverters
- Not for automotive applications unless otherwise specified

Features

- Extremely high CV product, ultra compact
- High reliability
- High ripple current capability
- RoHS-compatible

Construction

- Aluminum case, insulated with PET sleeve without insulation sheet at the can bottom
- Snap-in solder pins to hold component in place on PC-board
- Minus pole marking on shrinking sleeve
- Minus pole not insulated from case
- Overload protection by pressure relief device on the base

Terminals

- Standard version with 2 terminals,
2 lengths available: 6.3 and 4.5 mm
- 3 terminals to ensure correct insertion: length 4.5 mm



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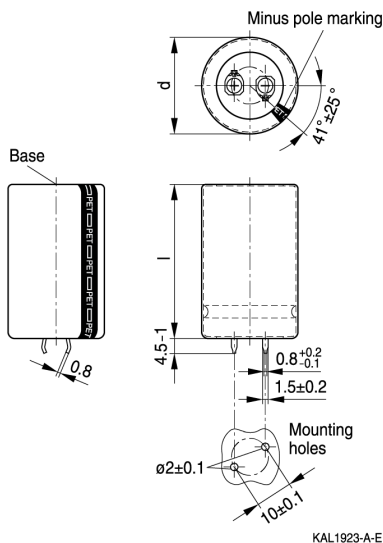
Specifications and characteristics in brief

Rated voltage V_R Surge voltage V_S	420 ... 500 V DC $1.10 \cdot V_R$ (15 ... 35 °C)		
Rated capacitance C_R Capacitance tolerance	80 ... 2060 μ F $\pm 20\% \triangleq M$		
Dissipation factor $\tan \delta$ (20 °C, 120 Hz)	≤ 0.20		
Leakage current I_{leak} (5 min, 20 °C)	$I_{leak} \leq 0.3 \mu A \cdot \left(\frac{C_R}{\mu F} \cdot \frac{V_R}{V} \right)^{0.7} + 4 \mu A$		
Self-inductance ESL	Approx. 20 nH		
Useful life ¹⁾ 105 °C; V_R ; $I_{AC,R}$	> 2000 h	Requirements: $ \Delta C/C \leq 20\%$ of initial value $\tan \delta \leq 2$ times initial specified limit $I_{leak} \leq$ initial specified limit	
Vibration resistance test	To IEC 60068-2-6:2007, test Fc: Frequency range 10 ... 55 Hz, displacement amplitude 0.35 mm, acceleration max. 5 g, duration 3 x 2 h. Capacitor mounted by its body which is rigidly clamped to the work surface.		
Characteristics at low temperature	Max. impedance ratio at 100 Hz	V_R	420 ... 500 V DC
		$Z_{-25\text{ °C}} / Z_{20\text{ °C}}$	10
		$Z_{-40\text{ °C}} / Z_{20\text{ °C}}$	20
IEC climatic category	To IEC 60068-1:2013: 25/105/56 (-25 °C/+105 °C/56 days damp heat test) The capacitors can be operated in the temperature range of –40 °C to +105 °C but the impedance at –40 °C must be taken into consideration.		
Sectional specification	IEC 60384-4:2016		

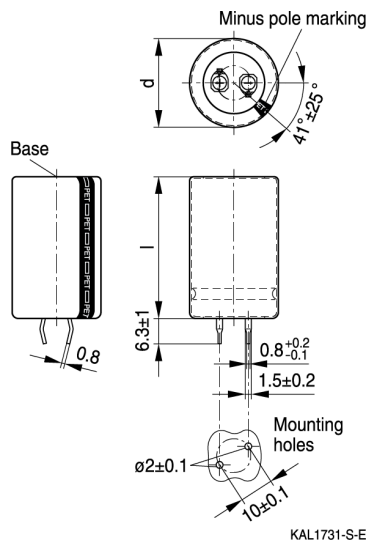
1) Refer to chapter "General technical information, 5 Useful life" on how to interpret useful life.

Dimensional drawings
Snap-in capacitors with PET shrinking sleeve

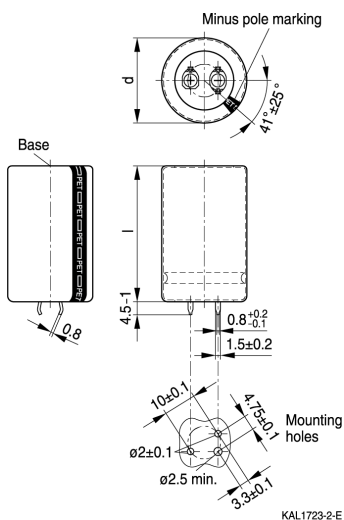
Insulation is marked with “PET” on the sleeve. Pressure relief device on the base.



Snap-in terminals (2 terminals, 4.5 mm)



Snap-in terminals (2 terminals, 6.3 mm)



Snap-in terminals (3 terminals, 4.5 mm)

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Dimensions and packing units

Dimensions (mm)		Approx. weight	Packing units
d +1	l + 2.5/-2	(g)	(pcs.)
22	25	9	160
22	30	12	160
22	35	15	160
22	40	18	160
22	45	20	160
22	50	24	160

Dimensions (mm)		Approx. weight	Packing units
d +1	l + 2.5/-2	(g)	(pcs.)
30	25	17	80
30	30	23	80
30	35	29	80
30	40	36	80
30	45	41	80
30	50	46	80
30	55	53	80
30	60	58	60
30	65	64	60
30	70	69	60
30	75	74	60
30	80	80	60
30	85	86	60
30	90	92	60
30	95	98	60
30	100	103	60

Dimensions (mm)		Approx. weight	Packing units
d +1	l + 2.5/-2	(g)	(pcs.)
25.4	25	13	130
25.4	30	17	130
25.4	35	19	130
25.4	40	22	130
25.4	45	25	130
25.4	50	29	130
25.4	55	32	130
25.4	60	36	45
25.4	65	39	45
25.4	70	42	45
25.4	75	45	45
25.4	80	48	45

Dimensions (mm)		Approx. weight	Packing units
d +1	l + 2.5/-2	(g)	(pcs.)
35	25	22	60
35	30	29	60
35	35	36	60
35	40	41	60
35	45	56	60
35	50	70	60
35	55	81	60
35	60	90	36
35	65	102	36
35	70	115	36
35	75	128	36
35	80	142	36
35	85	148	36
35	90	159	36
35	95	171	36
35	100	182	36

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Packaging of snap-in capacitors



For ecological reasons the packing is pure cardboard.

Ordering codes for terminal styles and insulation features

Identification in 3rd block of ordering code

Snap-in capacitors	
Terminal version	Insulation
	PET sleeve
Snap-in terminals (2 terminals, 4.5 mm)	M*57
Snap-in terminals (3 terminals, 4.5 mm)	M*52
Snap-in terminals (2 terminals, 6.3 mm)	M*50

* Digit used for nominal capacitance indication.

Ordering example:

- B43657C5917M057 } snap-in capacitor with 2 pin terminals (4.5 mm) and PET sleeve
B43657C5917M052 } snap-in capacitor with 3 pin terminals and PET sleeve

Overview of available types 1/4

The capacitance and voltage ratings listed below are available in different case sizes upon request. Other voltage and capacitance ratings are also available upon request.

V _R (V DC)	420	450	475	500
	Case dimensions d x l (mm)			
C _R (µF)				
80				22 x 25
110				22 x 30
120			22 x 25	25.4 x 25
130				22 x 35
140		22 x 25		
150	22 x 25			
160			22 x 30	22 x 40 25.4 x 30
170			25.4 x 25	
180		22 x 30		22 x 45
190		25.4 x 25	22 x 35	25.4 x 35 30 x 25
200	22 x 30			
210				22 x 50
220	25.4 x 25	22 x 35	25.4 x 30	
230			22 x 40	25.4 x 40
250	22 x 35	25.4 x 30		30 x 30
260		22 x 40	30 x 25	35 x 25
270			22 x 45 25.4 x 35	25.4 x 45
290	22 x 40 25.4 x 30	30 x 25		
300		25.4 x 35		25.4 x 50 30 x 35
310		22 x 45	22 x 50	
320			25.4 x 40	
330			30 x 30	30 x 35
340	22 x 45 30 x 25			25.4 x 55
350	25.4 x 35	22 x 50	35 x 25	
360		25.4 x 40		30 x 40
370			25.4 x 45	25.4 x 60
380	22 x 50	30 x 30		

Overview of available types 2/4

The capacitance and voltage ratings listed below are available in different case sizes upon request. Other voltage and capacitance ratings are also available upon request.

V _R (V DC)	420	450	475	500
	Case dimensions d x l (mm)			
C _R (μF)				
400		35 x 25		
410			30 x 35	30 x 45 35 x 35
420	25.4 x 40	25.4 x 45	25.4 x 50	
440	30 x 30			
450	35 x 25			
460			35 x 30	
470		30 x 35	25.4 x 55	30 x 50
480	25.4 x 45	25.4 x 50		35 x 40
490			30 x 40	
520		35 x 30	25.4 x 60	
530		25.4 x 55		30 x 55
540	25.4 x 50 30 x 35			
560		30 x 40	25.4 x 65 30 x 45	35 x 45
570			35 x 35	
580				30 x 60
590	35 x 30	25.4 x 60		
610	25.4 x 55		25.4 x 70	
620		25.4 x 65		
640	30 x 40	30 x 45 35 x 35	30 x 50	35 x 50
650			25.4 x 75	
670	25.4 x 60	25.4 x 70	35 x 40	
700			25.4 x 80	
710				35 x 55
720	35 x 35	25.4 x 75	30 x 55	
730		30 x 50		
740	30 x 45			
760		35 x 40		
780		25.4 x 80	35 x 45	
790				35 x 60

Overview of available types 3/4

The capacitance and voltage ratings listed below are available in different case sizes upon request. Other voltage and capacitance ratings are also available upon request.

V _R (V DC)	420	450	475	500
	Case dimensions d x l (mm)			
C _R (µF)				
800			30 x 60	
810			30 x 65	
820		30 x 55		
830	30 x 50			
860	35 x 40			
880		35 x 45	30 x 70 35 x 50	
910		30 x 60		
920		30 x 65		
930	30 x 55			
960			30 x 75	
990	35 x 45		35 x 55	
1000		35 x 50		
1010		30 x 70		
1030	30 x 60		30 x 80	
1090		30 x 75	35 x 60	
1100			30 x 85	
1110			35 x 65	
1120	35 x 50	35 x 55		
1170		30 x 80		
1180			30 x 90	
1210			35 x 70	
1250		35 x 60	30 x 95	
1260	35 x 55	30 x 85 35 x 65		
1310			35 x 75	
1320			30 x 100	
1340		30 x 90		
1380		35 x 70		
1390	35 x 60			
1410			35 x 80	
1420		30 x 95		

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Ultra compact – 105 °C
Overview of available types 4/4

The capacitance and voltage ratings listed below are available in different case sizes upon request. Other voltage and capacitance ratings are also available upon request.

V _R (V DC)	420	450	475	500
	Case dimensions d x l (mm)			
C _R (μF)				
1490		35 x 75		
1500		30 x 100		
1510			35 x 85	
1600		35 x 80		
1610			35 x 90	
1710			35 x 95	
1720		35 x 85		
1810			35 x 100	
1830		35 x 90		
1940		35 x 95		
2060		35 x 100		

Snap-in capacitors
B43657
Ultra compact – 105 °C
Technical data and ordering codes

C _R 100 Hz 20 °C μF	Case dimensions d x l mm	ESR _{typ} 120 Hz 20 °C mΩ	ESR _{typ} 360 Hz 60 °C mΩ	I _{AC,max} 120 Hz 60 °C A	I _{AC,max} 120 Hz 85 °C A	I _{AC,R} 120 Hz 105 °C A	Ordering code (composition see below)
V _R = 420 V DC							
150	22 x 25	860	210	2.00	1.51	0.86	B43657G0157M05#
200	22 x 30	650	160	2.53	1.90	1.09	B43657G0207M05#
220	25.4 x 25	600	160	2.53	1.91	1.09	B43657G0227M05#
250	22 x 35	520	130	3.04	2.29	1.31	B43657G0257M05#
290	22 x 40	450	110	3.44	2.59	1.48	B43657G0297M05#
290	25.4 x 30	460	120	3.16	2.38	1.36	B43657H0297M05#
340	22 x 45	380	95	3.95	2.98	1.70	B43657G0347M05#
340	30 x 25	410	120	3.18	2.39	1.37	B43657H0347M05#
350	25.4 x 35	380	100	3.69	2.78	1.59	B43657G0357M05#
380	22 x 50	340	85	4.35	3.27	1.87	B43657G0387M05#
420	25.4 x 40	320	85	4.29	3.23	1.85	B43657G0427M05#
440	30 x 30	320	100	3.89	2.93	1.67	B43657G0447M05#
450	35 x 25	330	120	3.53	2.66	1.52	B43657G0457M05#
480	25.4 x 45	280	70	4.81	3.62	2.07	B43657G0487M05#
540	25.4 x 50	240	65	5.33	4.01	2.29	B43657G0547M05#
540	30 x 35	260	80	4.59	3.45	1.97	B43657H0547M05#
590	35 x 30	250	90	4.33	3.26	1.74	B43657G0597M05#
610	25.4 x 55	220	55	5.92	4.45	2.55	B43657G0617M05#
640	30 x 40	220	65	5.27	3.97	2.12	B43657G0647M05#
670	25.4 x 60	200	50	6.43	4.84	2.59	B43657G0677M05#
720	35 x 35	210	75	5.07	3.81	2.04	B43657G0727M05#
740	30 x 45	190	55	5.96	4.48	2.39	B43657G0747M05#
830	30 x 50	170	50	6.58	4.95	2.64	B43657G0837M05#
860	35 x 40	170	65	5.85	4.40	2.35	B43657G0867M05#
930	30 x 55	150	45	7.25	5.46	2.92	B43657G0937M05#
990	35 x 45	150	55	6.57	4.94	2.64	B43657G0997M05#
1030	30 x 60	140	40	7.92	5.96	3.19	B43657G0108M35#
1120	35 x 50	130	50	7.30	5.49	2.93	B43657G0118M25#
1260	35 x 55	120	45	8.06	6.06	3.24	B43657G0128M65#
1390	35 x 60	110	40	8.78	6.60	3.53	B43657G0138M95#

Composition of ordering code

= Terminal style

7 = Snap-in terminals (2 terminals, 4.5 mm)

2 = Snap-in terminals (3 terminals, 4.5 mm)

0 = Snap-in terminals (2 terminals, 6.3 mm)

Snap-in capacitors
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Ultra compact – 105 °C
Technical data and ordering codes

C _R 120 Hz 20 °C μF	Case dimensions d x l mm	ESR _{typ} 120 Hz 20 °C mΩ	ESR _{typ} 360 Hz 60 °C mΩ	I _{AC,max} 120 Hz 60 °C A	I _{AC,max} 120 Hz 85 °C A	I _{AC,R} 120 Hz 105 °C A	Ordering code (composition see below)
V _R = 450 V DC							
140	22 x 25	830	220	2.01	1.52	0.90	B43657C5147M05#
180	22 x 30	650	170	2.46	1.86	1.10	B43657C5187M05#
190	25.4 x 25	630	170	2.41	1.82	1.07	B43657C5197M05#
220	22 x 35	530	140	2.91	2.20	1.30	B43657C5227M05#
250	25.4 x 30	480	130	3.00	2.26	1.34	B43657C5257M05#
260	22 x 40	450	120	3.35	2.53	1.49	B43657C5267M05#
290	30 x 25	430	130	3.00	2.27	1.34	B43657C5297M05#
300	25.4 x 35	400	110	3.48	2.63	1.55	B43657C5307M05#
310	22 x 45	380	100	3.89	2.94	1.73	B43657C5317M05#
350	22 x 50	330	85	4.32	3.26	1.92	B43657C5357M05#
360	25.4 x 40	330	90	4.05	3.06	1.81	B43657C5367M05#
380	30 x 30	330	100	3.71	2.79	1.65	B43657C5387M05#
400	35 x 25	330	120	3.42	2.58	1.52	B43657C5407M05#
420	25.4 x 45	280	80	4.62	3.48	2.06	B43657C5427M05#
470	30 x 35	270	85	4.40	3.31	1.96	B43657C5477M05#
480	25.4 x 50	250	70	5.19	3.91	2.31	B43657C5487M05#
520	35 x 30	260	100	4.17	3.14	1.70	B43657C5527M05#
530	25.4 x 55	220	60	5.66	4.27	2.52	B43657C5537M05#
560	30 x 40	220	70	5.07	3.82	2.07	B43657C5567M05#
590	25.4 x 60	200	55	6.21	4.68	2.54	B43657C5597M05#
620	25.4 x 65	210	55	6.47	4.87	2.60	B43657C5627M05#
640	30 x 45	200	60	5.68	4.28	2.33	B43657C5647M05#
640	35 x 35	210	80	4.91	3.70	2.01	B43657D5647M05#
670	25.4 x 70	190	50	6.94	5.23	2.79	B43657C5677M05#
720	25.4 x 75	180	45	7.41	5.58	2.98	B43657C5727M05#
730	30 x 50	170	55	6.35	4.79	2.60	B43657C5737M05#
760	35 x 40	170	65	5.64	4.25	2.31	B43657C5767M05#
780	25.4 x 80	170	45	7.95	5.99	3.20	B43657C5787M05#
820	30 x 55	150	50	7.02	5.29	2.87	B43657C5827M05#
880	35 x 45	150	55	6.36	4.79	2.60	B43657C5887M05#
910	30 x 60	140	45	7.68	5.78	3.14	B43657C5917M05#
920	30 x 65	150	45	8.15	6.12	3.26	B43657C5927M05#
1000	35 x 50	130	50	7.08	5.33	2.89	B43657C5108M05#
1010	30 x 70	130	40	8.81	6.61	3.53	B43657C5108M15#
1090	30 x 75	130	36	9.40	7.06	3.76	B43657C5108M95#
1120	35 x 55	120	45	7.79	5.87	3.18	B43657C5118M25#

Snap-in capacitors
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Ultra compact – 105 °C

C_R 120 Hz 20 °C μF	Case dimensions d x l mm	ESR_{typ} 120 Hz 20 °C m Ω	ESR_{typ} 360 Hz 60 °C m Ω	$I_{AC,max}$ 120 Hz 60 °C A	$I_{AC,max}$ 120 Hz 85 °C A	$I_{AC,R}$ 120 Hz 105 °C A	Ordering code (composition see below)
$V_R = 450$ V DC							
1170	30 x 80	120	34	9.99	7.50	4.00	B43657C5118M75#
1250	35 x 60	110	40	8.54	6.43	3.49	B43657C5128M55#
1260	30 x 85	110	32	10.60	7.99	4.26	B43657C5128M65#
1260	35 x 65	110	40	9.26	6.95	3.71	B43657D5128M65#
1340	30 x 90	100	30	11.20	8.43	4.50	B43657C5138M45#
1380	35 x 70	100	36	9.97	7.48	3.99	B43657C5138M85#
1420	30 x 95	100	28	11.80	8.87	4.73	B43657C5148M25#
1490	35 x 75	100	34	10.60	7.98	4.26	B43657C5148M95#
1500	30 x 100	90	26	12.40	9.31	4.97	B43657C5158M05#
1600	35 x 80	90	30	11.30	8.49	4.53	B43657C5168M05#
1720	35 x 85	85	30	12.00	9.01	4.81	B43657C5178M25#
1830	35 x 90	80	28	12.60	9.51	5.07	B43657C5188M35#
1940	35 x 95	75	26	13.30	10.00	5.34	B43657C5198M45#
2060	35 x 100	70	24	14.00	10.50	5.62	B43657C5208M65#

Composition of ordering code

= Terminal style

7 = Snap-in terminals (2 terminals, 4.5 mm)

2 = Snap-in terminals (3 terminals, 4.5 mm)

0 = Snap-in terminals (2 terminals, 6.3 mm)

Snap-in capacitors
B43657
Ultra compact – 105 °C
Technical data and ordering codes

C _R 120 Hz 20 °C μF	Case dimensions d x l mm	ESR _{typ} 120 Hz 20 °C mΩ	ESR _{typ} 360 Hz 60 °C mΩ	I _{AC,max} 120 Hz 60 °C A	I _{AC,max} 120 Hz 85 °C A	I _{AC,R} 120 Hz 105 °C A	Ordering code (composition see below)
V _R = 475 V DC							
120	22 x 25	1330	250	1.77	1.33	0.78	B43657C0127M05#
160	22 x 30	1000	190	2.23	1.68	0.99	B43657C0167M05#
170	25.4 x 25	960	190	2.20	1.66	0.98	B43657C0177M05#
190	22 x 35	840	160	2.57	1.94	1.14	B43657C0197M05#
220	25.4 x 30	740	150	2.71	2.04	1.20	B43657C0227M05#
230	22 x 40	700	130	3.02	2.27	1.34	B43657C0237M05#
260	30 x 25	650	140	2.79	2.10	1.24	B43657C0267M05#
270	22 x 45	590	110	3.46	2.61	1.54	B43657C0277M05#
270	25.4 x 35	600	120	3.20	2.41	1.42	B43657D0277M05#
310	22 x 50	520	100	3.91	2.94	1.73	B43657C0317M05#
320	25.4 x 40	510	100	3.68	2.77	1.63	B43657C0327M05#
330	30 x 30	510	110	3.36	2.53	1.49	B43657C0337M05#
350	35 x 25	490	130	3.20	2.41	1.42	B43657C0357M05#
370	25.4 x 45	440	85	4.17	3.14	1.85	B43657C0377M05#
410	30 x 35	410	90	4.00	3.01	1.78	B43657C0417M05#
420	25.4 x 50	390	75	4.65	3.50	2.06	B43657C0427M05#
460	35 x 30	380	100	3.92	2.95	1.60	B43657C0467M05#
470	25.4 x 55	350	70	5.14	3.86	2.28	B43657C0477M05#
490	30 x 40	340	75	4.62	3.48	1.89	B43657C0497M05#
520	25.4 x 60	310	60	5.61	4.22	2.29	B43657C0527M05#
560	25.4 x 65	310	60	6.12	4.60	2.46	B43657C0567M05#
560	30 x 45	300	65	5.18	3.90	2.12	B43657D0567M05#
570	35 x 35	300	80	4.64	3.49	1.89	B43657C0577M05#
610	25.4 x 70	280	55	6.60	4.96	2.65	B43657C0617M05#
640	30 x 50	260	60	5.80	4.36	2.37	B43657C0647M05#
650	25.4 x 75	260	50	7.00	5.26	2.81	B43657C0657M05#
670	35 x 40	260	70	5.29	3.98	2.16	B43657C0677M05#
700	25.4 x 80	250	50	7.48	5.63	3.01	B43657C0707M05#
720	30 x 55	230	55	6.41	4.82	2.62	B43657C0727M05#
780	35 x 45	220	60	5.98	4.50	2.44	B43657C0787M05#
800	30 x 60	210	50	7.03	5.29	2.87	B43657C0807M05#
810	30 x 65	210	45	7.73	5.8	3.10	B43657C0817M05#
880	30 x 70	200	45	8.29	6.22	3.32	B43657C0887M05#
880	35 x 50	200	55	6.63	4.99	2.71	B43657D0887M05#
960	30 x 75	180	40	8.91	6.69	3.57	B43657C0967M05#
990	35 x 55	180	50	7.32	5.50	2.99	B43657C0997M05#

Snap-in capacitors
B43657
Ultra compact – 105 °C

C _R 120 Hz 20 °C μF	Case dimensions d x l mm	ESR _{typ} 120 Hz 20 °C mΩ	ESR _{typ} 360 Hz 60 °C mΩ	I _{AC,max} 120 Hz 60 °C A	I _{AC,max} 120 Hz 85 °C A	I _{AC,R} 120 Hz 105 °C A	Ordering code (composition see below)
V _R = 475 V DC							
1030	30 x 80	170	38	9.47	7.11	3.79	B43657C0108M35#
1090	35 x 60	160	45	7.95	5.98	3.25	B43657C0108M95#
1100	30 x 85	160	34	10.00	7.53	4.02	B43657C0118M05#
1110	35 x 65	160	40	8.87	6.65	3.55	B43657C0118M15#
1180	30 x 90	150	32	10.60	7.99	4.26	B43657C0118M85#
1210	35 x 70	150	40	9.52	7.15	3.81	B43657C0128M15#
1250	30 x 95	140	30	11.20	8.40	4.48	B43657C0128M55#
1310	35 x 75	140	36	10.10	7.63	4.07	B43657C0138M15#
1320	30 x 100	130	30	11.70	8.82	4.71	B43657C0138M25#
1410	35 x 80	130	32	10.80	8.12	4.33	B43657C0148M15#
1510	35 x 85	120	30	11.40	8.61	4.59	B43657C0158M15#
1610	35 x 90	110	30	12.10	9.09	4.85	B43657C0168M15#
1710	35 x 95	110	28	12.70	9.57	5.11	B43657C0178M15#
1810	35 x 100	100	26	13.40	10.00	5.36	B43657C0188M15#

Composition of ordering code

= Terminal style

7 = Snap-in terminals (2 terminals, 4.5 mm)

2 = Snap-in terminals (3 terminals, 4.5 mm)

0 = Snap-in terminals (2 terminals, 6.3 mm)

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Technical data and ordering codes

C _R 100 Hz 20 °C μF	Case dimensions d x l mm	ESR _{typ} 120 Hz 20 °C mΩ	ESR _{typ} 360 Hz 60 °C mΩ	I _{AC,max} 120 Hz 60 °C A	I _{AC,max} 120 Hz 85 °C A	I _{AC,R} 120 Hz 105 °C A	Ordering code (composition see below)
V _R = 500 V DC							
80	22 x 25	1150	310	1.48	1.12	0.56	B43657C6806M05#
110	22 x 30	840	230	1.91	1.44	0.72	B43657C6117M05#
120	25.4 x 25	780	220	1.94	1.46	0.74	B43657C6127M05#
130	22 x 35	710	190	2.19	1.66	0.83	B43657C6137M05#
160	22 x 40	580	160	2.62	1.97	0.99	B43657C6167M05#
160	25.4 x 30	580	160	2.43	1.83	0.92	B43657D6167M05#
180	22 x 45	510	140	2.89	2.18	1.10	B43657C6187M05#
190	25.4 x 35	490	140	2.80	2.11	1.06	B43657C6197M05#
190	30 x 25	510	150	2.58	1.94	0.98	B43657D6197M05#
210	22 x 50	440	120	3.31	2.49	1.25	B43657C6217M05#
230	25.4 x 40	410	110	3.28	2.48	1.24	B43657C6237M05#
250	30 x 30	390	120	3.18	2.40	1.21	B43657C6257M05#
260	35 x 25	390	130	3.05	2.30	1.16	B43657C6267M05#
270	25.4 x 45	350	100	3.76	2.83	1.42	B43657C6277M05#
300	25.4 x 50	310	85	4.12	3.10	1.56	B43657C6307M05#
300	30 x 35	320	100	3.69	2.78	1.40	B43657D6307M05#
330	35 x 30	300	100	3.66	2.76	1.31	B43657C6337M05#
340	25.4 x 55	270	75	4.59	3.46	1.74	B43657C6347M05#
360	30 x 40	270	80	4.28	3.22	1.53	B43657C6367M05#
370	25.4 x 60	250	70	4.95	3.73	1.77	B43657C6377M05#
410	30 x 45	230	70	4.77	3.59	1.70	B43657C6417M05#
410	35 x 35	250	85	4.34	3.27	1.55	B43657D6417M05#
470	30 x 50	210	65	5.35	4.03	1.91	B43657C6477M05#
480	35 x 40	210	70	4.92	3.71	1.76	B43657C6487M05#
530	30 x 55	180	55	5.93	4.47	2.12	B43657C6537M05#
560	35 x 45	180	60	5.58	4.20	1.99	B43657C6567M05#
580	30 x 60	170	50	6.42	4.84	2.30	B43657C6587M05#
640	35 x 50	160	55	6.23	4.69	2.22	B43657C6647M05#
710	35 x 55	140	50	6.81	5.13	2.43	B43657C6717M05#
790	35 x 60	130	45	7.45	5.61	2.66	B43657C6797M05#

Composition of ordering code

= Terminal style

7 = Snap-in terminals (2 terminals, 4.5 mm)

2 = Snap-in terminals (3 terminals, 4.5 mm)

0 = Snap-in terminals (2 terminals, 6.3 mm)

Useful life¹⁾

For useful life calculations, please use our web-based "AlCap Useful Life Calculation Tool", which can be found on the Internet under the following link:

www.tdk-electronics.tdk.com/alcap

The AlCap Useful Life Calculation Tool provides calculations of useful life as well as additional data for selected capacitor types under operating conditions defined by the user.

In addition, it is possible to calculate useful life expectancies based on temperatures measured by the user in the application.

1) Refer to chapter "General technical information, 5 Useful life" on how to interpret useful life."

Cautions and warnings

Personal safety

The electrolytes used have been optimized both with a view to the intended application and with regard to health and environmental compatibility.

As far as possible, we do not use any dangerous chemicals or compounds to produce operating electrolytes, although in exceptional cases, such materials must be used in order to achieve specific physical and electrical properties because no alternative materials are currently known.

We do, however, restrict the amount of dangerous materials used in our products to an absolute minimum.

Materials and chemicals used in our Aluminum Electrolytic Capacitors and Hybrid Polymer Capacitors are continuously adapted in compliance with the TDK Electronics Corporate Environmental Policy and the latest EU regulations and guidelines such as RoHS, REACH/SVHC, GADSL, and ELV.

MDS (Material Data Sheets) are available on our website for types listed on the product information site of TDK.

MDS for customer specific capacitors are available upon request.

Nevertheless, the following rules should be observed when handling Aluminum Electrolytic Capacitors and Hybrid Polymer Capacitors: No electrolyte should come into contact with eyes or skin. If electrolyte does come into contact with the skin, wash the affected areas immediately with running water. If the eyes are affected, rinse them for 10 minutes with plenty of water. If symptoms persist, seek medical treatment. Avoid inhaling electrolyte vapor or mists. Workplaces and other affected areas should be well ventilated. Clothing that has been contaminated by electrolyte must be changed and rinsed in water.

Product safety

The table below summarizes the safety instructions that must be observed without fail. A detailed description can be found in the relevant sections of separate file chapter “General technical information”.

Topic	Safety information	Reference chapter “General technical information”
Polarity	Make sure that polar capacitors are connected with the right polarity.	1 “Basic construction of aluminum electrolytic capacitors”
Reverse voltage	Voltages of opposite polarity should be prevented by connecting a diode.	3.1.6 “Reverse voltage”
Operating temperature range	Do not exceed the temperature limits specified in the product data sheet. Useful life and reliability depend to a large extent on the capacitor's temperature.	7.1, 7.2 “Minimum and maximum permissible operating temperatures”
Storage	Do not store capacitors at high temperatures or high humidity. Capacitors should be stored at +5 to +35 °C and a relative humidity of $\leq 75\%$.	7.3 “Shelf life and storage conditions”
Mounting of capacitors	The internal and external structure of capacitors might be damaged if excessive force is applied to their terminals. Avoid any compressive, tensile or flexural stress. Do not move the capacitor after soldering to PC board. Do not pick up the PC board by the soldered capacitor. Do not insert the capacitor on the PC board with a hole space different to the lead space specified.	11 “Mounting”
Mounting position of capacitors	The position of the pressure relief device must be taken into consideration when designing the application layout. As a rule, the function of the pressure relief device must not be impaired or blocked by the mounting. Screw terminal capacitors, with pressure relief device in the cover disk, must not be mounted with terminals facing down unless otherwise specified. Multi-pin capacitors with pressure relief device on the can base must not be mounted with terminals facing up unless otherwise specified.	11.1 “Mounting positions of capacitors”

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Topic	Safety information	Reference chapter “General technical information”
Robustness of terminals	Unless otherwise specified, the following maximum tightening torques must not be exceeded when connecting screw terminals: M5: 2.5 Nm M6: 4.0 Nm	11.2 “Mounting torques”
Potting and gluing	If applied, potting and gluing materials must not impair the function of the capacitor's pressure relief device.	11.4 “Potting and gluing”
Soldering	Do not exceed the specified time or temperature limits during soldering.	11.5 “Soldering”
Soldering, cleaning agents	Do not allow halogenated hydrocarbons to come into contact with Aluminum Electrolytic Capacitors or Hybrid Polymer Capacitors.	11.6 “Cleaning agents”
Passive flammability	Avoid external energy, e.g. fire.	8.1 “Passive flammability”
Active flammability	Avoid overload of the capacitors.	8.2 “Active flammability”
Maintenance	Make periodic inspections of the capacitors. Before the inspection, make sure that the power supply is turned off and carefully discharge the capacitors. Do not apply excessive mechanical stress to the capacitor terminals when mounting.	10 “Maintenance”
		Reference chapter “Capacitors with screw terminals”
Breakdown strength of shrinking sleeve	Do not damage the shrinking sleeve, especially when ring clips are used for mounting.	“Screw terminals – accessories”

Display of ordering codes for TDK Electronics products

The ordering code for one and the same product can be represented differently in data sheets, data books, other publications, on the company website, or in order-related documents such as shipping notes, order confirmations and product labels. The varying representations of the ordering codes are due to different processes employed and do not affect the specifications of the respective products.

Detailed information can be found on the Internet under www.tdk-electronics.tdk.com/orderingcodes.

Symbols and terms

Symbol	English	German
C	Capacitance	Kapazität
ΔC	Capacitance difference	Kapazitätsdifferenz
C_R	Rated capacitance	Nennkapazität
C_S	Series capacitance	Serienkapazität
$C_{S,T}$	Series capacitance at temperature T	Serienkapazität bei Temperatur T
C_f	Capacitance at frequency f	Kapazität bei Frequenz f
d	Case diameter, nominal dimension	Gehäusedurchmesser, Nennmaß
d_{max}	Maximum case diameter	Maximaler Gehäusedurchmesser
ESL	Self-inductance	Eigeninduktivität
ESR	Equivalent series resistance	Ersatzserienwiderstand
ESR_f	Equivalent series resistance at frequency f	Ersatzserienwiderstand bei Frequenz f
ESR_T	Equivalent series resistance at temperature T	Ersatzserienwiderstand bei Temperatur T
f	Frequency	Frequenz
I	Current	Strom
I_{AC}	Alternating current (ripple current)	Wechselstrom
$I_{AC,RMS}$	Root-mean-square value of alternating current	Wechselstrom, Effektivwert
$I_{AC,f}$	Ripple current at frequency f	Wechselstrom bei Frequenz f
$I_{AC,max}$	Maximum permissible ripple current	Maximal zulässiger Wechselstrom
$I_{AC,R}$	Rated ripple current	Nennwechselstrom
I_{leak}	Leakage current	Reststrom
$I_{leak,op}$	Operating leakage current	Betriebsreststrom
l	Case length, nominal dimension	Gehäuselänge, Nennmaß
l_{max}	Maximum case length (without terminals and mounting stud)	Maximale Gehäuselänge (ohne Anschlüsse und Gewindebolzen)
R	Resistance	Widerstand
R_{ins}	Insulation resistance	Isolationswiderstand
R_{symm}	Balancing resistance	Symmetrierwiderstand
T	Temperature	Temperatur
ΔT	Temperature difference	Temperaturdifferenz
T_A	Ambient temperature	Umgebungstemperatur
T_B	Capacitor base temperature	Temperatur des Gehäusebodens
T_C	Case temperature	Gehäusetemperatur
t	Time	Zeit
Δt	Period	Zeitraum
t_b	Service life (operating hours)	Brauchbarkeitsdauer (Betriebszeit)
V	Voltage	Spannung
V_F	Forming voltage	Formierspannung
V_{op}	Operating voltage	Betriebsspannung
V_R	Rated voltage, DC voltage	Nennspannung, Gleichspannung
V_S	Surge voltage	Spitzenspannung

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Symbol	English	German
X_C	Capacitive reactance	Kapazitiver Blindwiderstand
X_L	Inductive reactance	Induktiver Blindwiderstand
Z	Impedance	Scheinwiderstand
Z_T	Impedance at temperature T	Scheinwiderstand bei Temperatur T
$\tan \delta$	Dissipation factor	Verlustfaktor
λ	Failure rate	Ausfallrate
ϵ_0	Absolute permittivity	Elektrische Feldkonstante
ϵ_r	Relative permittivity	Dielektrizitätszahl
ω	Angular frequency; $2 \cdot \pi \cdot f$	Kreisfrequenz; $2 \cdot \pi \cdot f$

Note:

All dimensions are given in mm.

Important notes

The following applies to all products named in this publication:

1. Some parts of this publication contain **statements about the suitability of our products for certain areas of application**. These statements are based on our knowledge of typical requirements that are often placed on our products in the areas of application concerned. We nevertheless expressly point out **that such statements cannot be regarded as binding statements about the suitability of our products for a particular customer application**. As a rule we are either unfamiliar with individual customer applications or less familiar with them than the customers themselves. For these reasons, it is always ultimately incumbent on the customer to check and decide whether a product with the properties described in the product specification is suitable for use in a particular customer application.
2. We also point out that **in individual cases, a malfunction of electronic components or failure before the end of their usual service life cannot be completely ruled out in the current state of the art, even if they are operated as specified**. In customer applications requiring a very high level of operational safety and especially in customer applications in which the malfunction or failure of an electronic component could endanger human life or health (e.g. in accident prevention or life-saving systems), it must therefore be ensured by means of suitable design of the customer application or other action taken by the customer (e.g. installation of protective circuitry or redundancy) that no injury or damage is sustained by third parties in the event of malfunction or failure of an electronic component.
3. **The warnings, cautions and product-specific notes must be observed.**
4. In order to satisfy certain technical requirements, **some of the products described in this publication may contain substances subject to restrictions in certain jurisdictions (e.g. because they are classed as hazardous)**. Useful information on this will be found in our Material Data Sheets on the Internet (www.tdk-electronics.tdk.com/material). Should you have any more detailed questions, please contact our sales offices.
5. We constantly strive to improve our products. Consequently, **the products described in this publication may change from time to time**. The same is true of the corresponding product specifications. Please check therefore to what extent product descriptions and specifications contained in this publication are still applicable before or when you place an order.
We also **reserve the right to discontinue production and delivery of products**. Consequently, we cannot guarantee that all products named in this publication will always be available. The aforementioned does not apply in the case of individual agreements deviating from the foregoing for customer-specific products.
6. Unless otherwise agreed in individual contracts, **all orders are subject to our General Terms and Conditions of Supply**.

Important notes

7. **Our manufacturing sites serving the automotive business apply the IATF 16949 standard.**
The IATF certifications confirm our compliance with requirements regarding the quality management system in the automotive industry. Referring to customer requirements and customer specific requirements ("CSR") TDK always has and will continue to have the policy of respecting individual agreements. Even if IATF 16949 may appear to support the acceptance of unilateral requirements, we hereby like to emphasize that **only requirements mutually agreed upon can and will be implemented in our Quality Management System**. For clarification purposes we like to point out that obligations from IATF 16949 shall only become legally binding if individually agreed upon.
8. The trade names EPCOS, CarXield, CeraCharge, CeraDiode, CeraLink, CeraPad, CeraPlas, CSMP, CTVS, DeltaCap, DigiSiMic, FilterCap, FormFit, InsuGate, LeaXield, MediPlas, MiniBlue, MiniCell, MKD, MKK, ModCap, MotorCap, PCC, PhaseCap, PhaseCube, PhaseMod, PhiCap, PiezoBrush, PlasmaBrush, PowerHap, PQSine, PQvar, SIFERRIT, SIFI, SIKOREL, SilverCap, SIMDAD, SiMic, SIMID, SineFormer, SIOV, SurfIND, ThermoFuse, WindCap, XieldCap are **trademarks registered or pending** in Europe and in other countries. Further information will be found on the Internet at www.tdk-electronics.tdk.com/trademarks.

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